

Week 3: Inequalities

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Week 3 Quiz

3.1: Inequalities with absolute value; solution sets & graphs

3.2: Literal equations & rearranging formulas (solve for a variable)

3.3: Absolute-value equations; modeling simple contexts

Quick Reference: Equations, Definitions, & Formulas

S.A.D.D. — Signs • Arithmetic • Distribute • Denominators

- **Signs:** When adding or subtracting, keep the sign of the larger absolute value.
- **Arithmetic:** Follow order of operations (PEMDAS).
- **Distribute:** Use the distributive property to eliminate parentheses.
- **Denominators:** Multiply both sides by the least common denominator to eliminate fractions.

Units?

Test a value

Inequality Symbols

- $<$ means "less than"
- $\leq$ means "less than or equal to"
- $>$ means "greater than"
- $\geq$ means "greater than or equal to"

Solving Linear Inequalities

- Solve like an equation, but remember:
- **When multiplying or dividing by a negative number, flip the inequality sign!**
- Example: If $-2x < 6$, divide by -2 : $x > -3$ (sign flipped)

Compound Inequalities

AND: Both conditions must be true (intersection)

Example: $-2 \leq x < 5$ means x is between -2 and 5

OR: At least one condition must be true (union)

Example: $x < -1$ OR $x > 3$ means x is less than -1 or greater than 3

Absolute Value Inequalities

For $|x| < a$ (where $a > 0$): $-a < x < a$

For $|x| > a$ (where $a > 0$): $x < -a$ OR $x > a$

Interval Notation

- (a, b) means $a < x < b$ (open interval, excludes endpoints)
- $[a, b]$ means $a \leq x \leq b$ (closed interval, includes endpoints)
- $[a, b)$ means $a \leq x < b$ (half-open interval)
- $(-\infty, a)$ means $x < a$
- $[a, \infty)$ means $x \geq a$

Week 3 Quiz: Inequalities

Instructions: Show all work. Circle your final answer.

Part A: Linear Inequalities

1) Solve the inequality: $2x - 5 \geq 9$.

- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

2) Solve the inequality: $-3x + 4 < 1$.

- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

Part B: Compound Inequalities

3) Solve the compound inequality: $-2 \leq x + 1 < 5$.

- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

4) Solve the compound inequality: $2x - 5 \geq 9$ **OR** $x + 3 < -1$.

- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

Part C: Absolute Value Inequalities

5) Solve the inequality: $|x - 3| < 5$.

- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

6) Solve the inequality: $|2x + 1| \geq 7$.

- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

Part D: Word Problems

7) A rideshare charges a \$3 base fee plus \$0.60 per mile.

You have at most \$15 to spend.

Let d = miles traveled (in miles).

Assume $d \geq 0$.

- 1. Write and solve an inequality for d .
- 2. State the greatest whole number of miles you can travel.
- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

8) A resistor is labeled $220\ \Omega$ (ohms).

It is acceptable if the resistance differs by at most $5\ \Omega$.

Let R = measured resistance (in ohms).

- 1. Write and solve an inequality for R .
- 2. State the acceptable range.
- (a) Graph the solution set on a number line.
- (b) Write the solution in interval notation.

Week 2 Challenge Review Questions

Solve the equation: $\frac{1}{3}(x - 2) = 0.25x + 7$

Given the equation: $ax + b = cx + d$, describe the solution set for each case:

Case 1: $a = c$ AND $b = d$

Case 2: $a = c$ AND $b \neq d$

Case 3: $a \neq c$ AND $b = d$

Case 4: $a \neq c$ AND $b \neq d$

Solution Guide

Part A: Linear Inequalities

1. Solve the inequality: $2x - 5 \geq 9$.

Solution:

Add 5 to both sides:

$$2x - 5 + 5 \geq 9 + 5$$

$$2x \geq 14$$

Divide by 2:

$$x \geq 7$$

(a) *Graph:* Closed circle at 7, shaded to the right $\rightarrow$

(b) *Interval notation:* $\boxed{[7, \infty)}$

2. Solve the inequality: $-3x + 4 < 1$.

Solution:

Subtract 4 from both sides:

$$-3x + 4 - 4 < 1 - 4$$

$$-3x < -3$$

Divide by -3 (flip the inequality sign!):

$$x > 1$$

(a) *Graph:* Open circle at 1, shaded to the right $\rightarrow$

(b) *Interval notation:* $\boxed{(1, \infty)}$

Part B: Compound Inequalities

3. Solve the compound inequality: $-2 \leq x + 1 < 5$.

Solution:

Subtract 1 from all three parts:

$$-2 - 1 \leq x + 1 - 1 < 5 - 1$$

$$-3 \leq x < 4$$

(a) *Graph:* Closed circle at -3 , open circle at 4 , shaded between

(b) *Interval notation:* $\boxed{[-3, 4)}$

4. Solve the compound inequality: $2x - 5 \geq 9$ OR $x + 3 < -1$.

Solution:

Solve each inequality separately:

First inequality: $2x - 5 \geq 9$

$$2x \geq 14$$

$$x \geq 7$$

Second inequality: $x + 3 < -1$

$$x < -4$$

Solution: $x < -4$ OR $x \geq 7$

(a) *Graph:* Open circle at -4 , shaded left $\leftarrow$; Closed circle at 7 , shaded right $\rightarrow$

(b) *Interval notation:* $\boxed{(-\infty, -4) \cup [7, \infty)}$

Part C: Absolute Value Inequalities

5. Solve the inequality: $|x - 3| < 5$.

Solution:

For $|x - 3| < 5$, we have:

$$-5 < x - 3 < 5$$

Add 3 to all parts:

$$-5 + 3 < x < 5 + 3$$

$$-2 < x < 8$$

(a) *Graph:* Open circles at -2 and 8 , shaded between

(b) *Interval notation:* $\boxed{(-2, 8)}$

6. Solve the inequality: $|2x + 1| \geq 7$.

Solution:

For $|2x + 1| \geq 7$, we have:

$$2x + 1 \leq -7 \text{ OR } 2x + 1 \geq 7$$

First inequality: $2x + 1 \leq -7$

$$2x \leq -8$$

$$x \leq -4$$

Second inequality: $2x + 1 \geq 7$

$$2x \geq 6$$

$$x \geq 3$$

Solution: $x \leq -4$ OR $x \geq 3$

(a) *Graph:* Closed circle at -4 , shaded left $\leftarrow$; Closed circle at 3 , shaded right $\rightarrow$

(b) *Interval notation:* $\boxed{(-\infty, -4] \cup [3, \infty)}$

Part D: Word Problems

7. WP1) A rideshare charges a \$3 base fee plus \$0.60 per mile. You have at most \$15 to spend.

Solution:

Set up the inequality:

Cost = base fee + (cost per mile $\times$ miles)

$$3 + 0.60d \leq 15$$

Subtract 3 from both sides:

$$0.60d \leq 12$$

Divide by 0.60:

$$d \leq 20$$

Since $d \geq 0$, the solution is $0 \leq d \leq 20$

Greatest whole number of miles: 20 miles

(a) *Graph:* Closed circles at 0 and 20, shaded between

(b) *Interval notation:* $[0, 20]$

8. WP2) A resistor is labeled 220Ω (ohms). It is acceptable if the resistance differs by at most 5Ω .

Solution:

The resistance can differ by at most 5Ω , so:

$$|R - 220| \leq 5$$

This means:

$$-5 \leq R - 220 \leq 5$$

Add 220 to all parts:

$$215 \leq R \leq 225$$

Acceptable range: 215Ω to 225Ω

(a) *Graph:* Closed circles at 215 and 225, shaded between

(b) *Interval notation:* $[215, 225]$

Week 2 Challenge Review Questions

Solve the equation: $\frac{1}{3}(x - 2) = 0.25x + 7$

Solution:

Note that $\frac{1}{4} = 0.25$, so:

$$0.25(x - 2) = 0.25x + 7$$

Multiply both sides by 3:

$$x - 2 = 0.75x + 21$$

Subtract $0.75x$ from both sides:

$$0.25x - 2 = 21$$

Add $2x$ to both sides:

$$0.25x = 23$$

Note that $\frac{1}{4} = 0.25$, so:

$$\frac{1}{4}x = 23$$

Multiply both sides by 4:

$$x = 92 \text{ This is a false statement!}$$

Answer: No solution (contradiction)

Given the equation: $ax + b = cx + d$, describe the solution set for each case:

Case 1: $a = c$ AND $b = d$

Solution:

The equation becomes $ax + b = ax + b$, which is always true.

Answer: All real numbers (identity)

Case 2: $a = c$ AND $b \neq d$

Solution:

The equation becomes $ax + b = ax + d$ where $b \neq d$.

Subtracting ax from both sides gives $b = d$, which contradicts $b \neq d$.

Answer: No solution (contradiction)

Case 3: $a \neq c$ AND $b = d$

Solution:

The equation becomes $ax + b = cx + b$.

Subtracting b from both sides: $ax = cx$
Rearranging: $(a - c)x = 0$
Since $a \neq c$, we have $a - c \neq 0$, so $x = 0$.

Answer: One solution, $x = 0$

Case 4: $a \neq c$ AND $b \neq d$

Solution:

Rearranging: $ax - cx = d - b$

$(a - c)x = d - b$

Since $a \neq c$, we can divide: $x = \frac{d-b}{a-c}$

Answer: One solution, $x = \frac{d-b}{a-c}$
